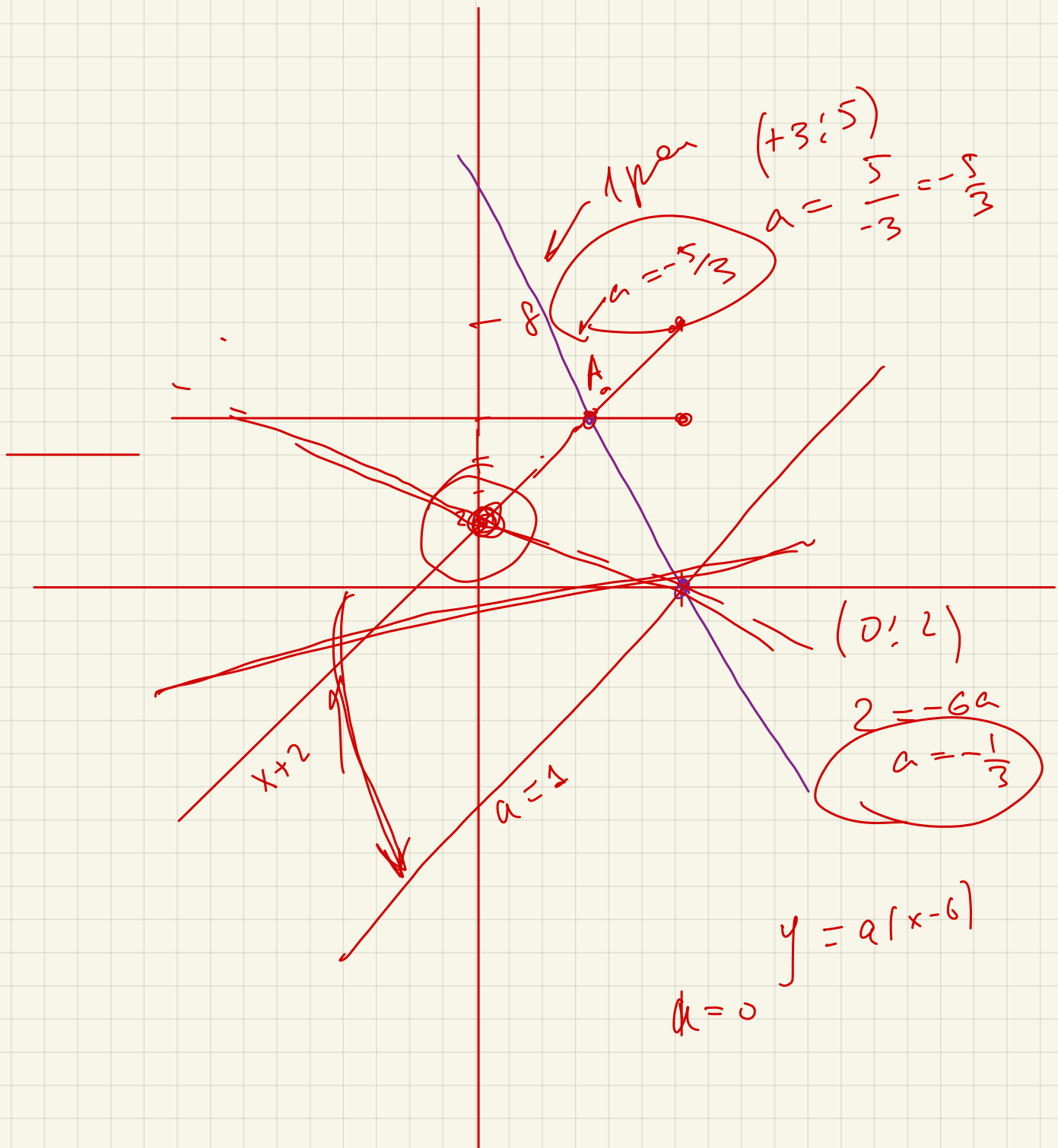






$$\left(x - \frac{1}{4}\right)^2 + \left(y - \frac{1}{4}\right)^2 = \frac{1}{8}$$

$$x^2 - \frac{x}{2} + \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

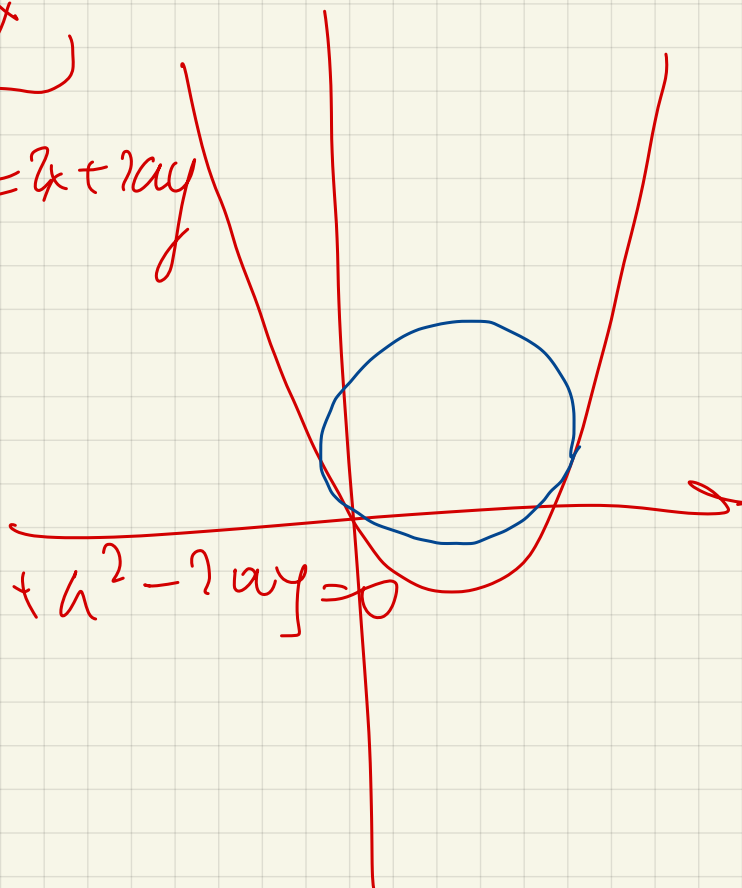
$$x(2x-1) = 0$$

$$x = 0$$

$$x = \frac{1}{2}$$

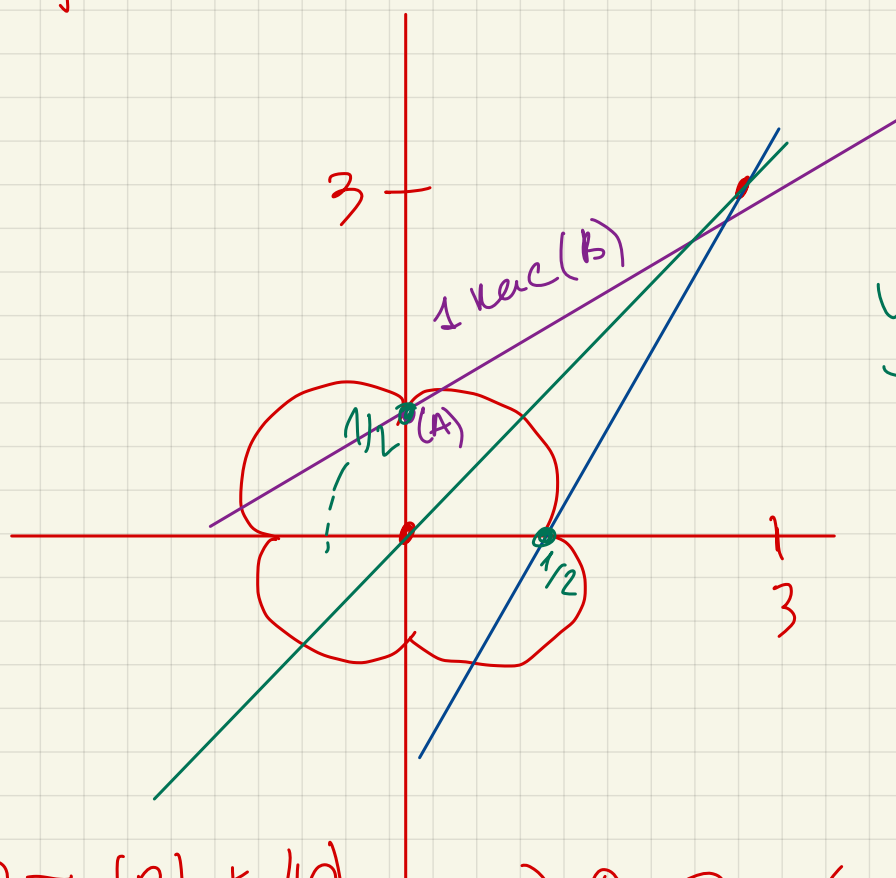
$$\frac{2x^2 - x}{2} = 0$$

$$\begin{cases} y = x^2 - 2x \\ x^2 + y^2 + a^2 = 2x + 2ay \end{cases}$$



$$x^2 - 2x + y^2 + a^2 - 2ay = 0$$

$$y^2 + y + a^2 - 2ay = 0 \quad D = 0$$



$$\begin{aligned} y &= a(x-3)+3 \\ \frac{1}{2} &= 3a+3 \\ \frac{1}{2}-3 & \\ \frac{1}{2}-\frac{9}{2} &= -3a \\ a &= \frac{5}{6} \end{aligned}$$

$$0 + 0 = |0| + |0| \rightarrow 0 = 0 \quad \checkmark$$

$$(a(x-3)+3)(a(x-3)+3) =$$

$$a^2(x-3)^2 + 3 \cdot 2 \cdot a(x-3) + 9 =$$

$$a^2(x^2 - 6x + 9) + 6ax - 18a + 9$$

$$\left(a(x-3) + 3 - \frac{1}{4} \right) \left(a(x-3) + \frac{11}{4} \right)^2$$

$$x^2 - 2 \cdot \frac{1}{4} \cdot x + \left(\frac{1}{4} \right)^2 + a^2(x-3)^2 + 2 \cdot \frac{11}{4} a(x-3) + \left(\frac{11}{4} \right)^2 =$$

$$x^2 + a^2 x^2 - \frac{x}{2} + \frac{1}{16} + a^2 \cdot 9 - 6a^2 x + \frac{11ax}{2} - \frac{33a}{2} + \frac{121}{16} = \frac{1}{8}$$

$$x^2(1+a^2) - x\left(\frac{1}{2} - 6a^2 + \frac{11a}{2}\right) + \frac{1}{16} + \frac{121}{16} - \frac{1}{8} + 9a^2 - \frac{33a}{2} = \frac{15}{2}$$

$$x^2(1+a^2) - x\left(-6a^2 + \frac{11a}{2} + \frac{1}{2}\right) + 9a^2 - \frac{33a}{2} + \frac{15}{2} =$$

$$2x^2(1+a^2) - x(-12a^2+11a+1) + 18a^2 - 33a + 5$$

$$(-12a^2+11a+1)^2 - 2 \cdot 2(1+a^2)(18a^2-33a+5)$$

$$\log_{2023} \left(1012 + 1011 \frac{|x|}{x} \right) + (x-y)^2 = a$$

$$(2) \log_{2024}(3-y) = 0 \leftarrow$$

$$a \rightarrow 2 \text{ pen}$$

$$y < 3 \Rightarrow 3-y = 2024^0 = 1$$

$$\underline{y = 2}$$

$$\log_{2023} \left(1012 + 1011 \frac{|x|}{x} \right) + (x-2)^2 = a$$

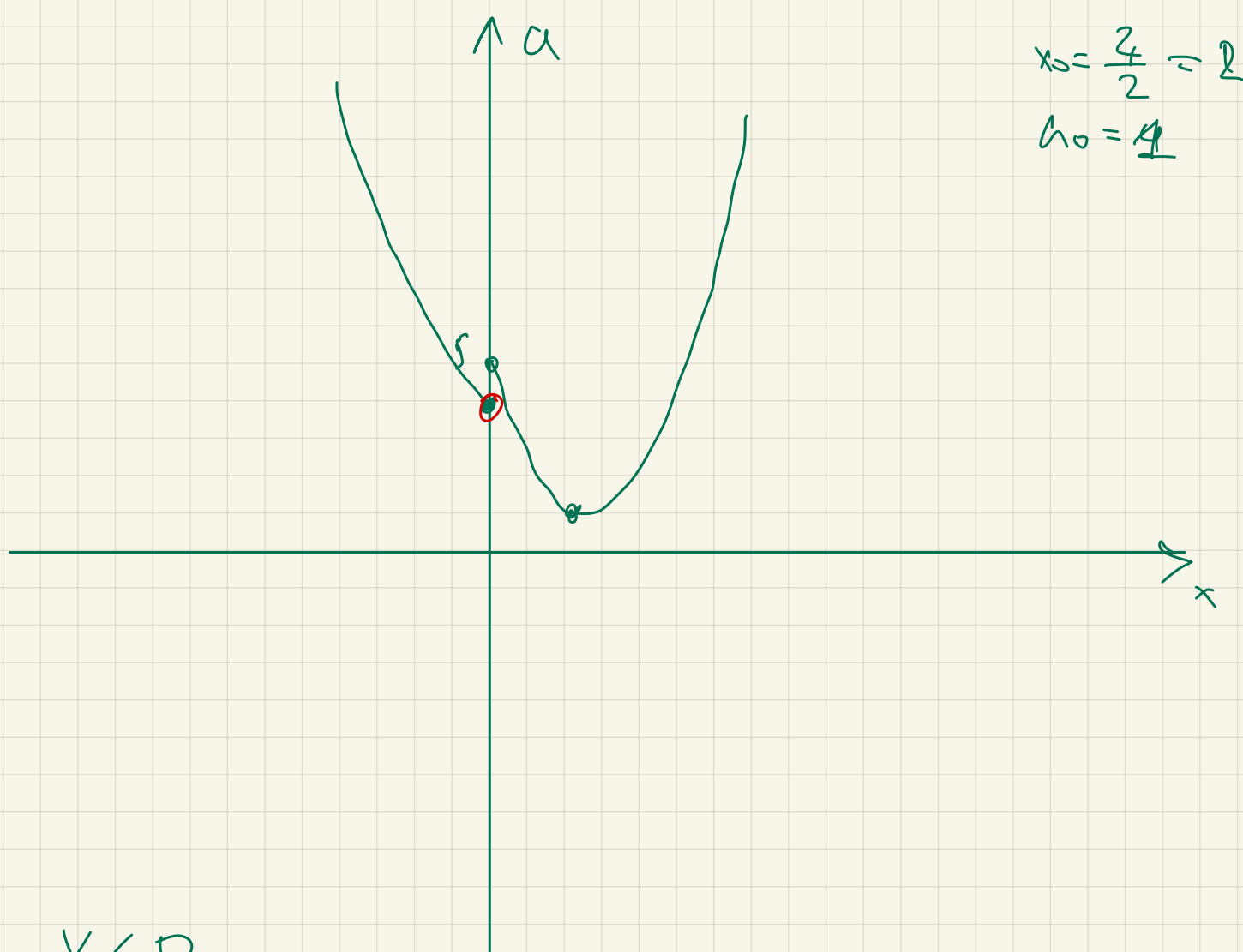
$$x \geq 0 \quad \log_{2023} (1012 + 1011) + (x-2)^2 = a$$

$$\log_{2023} 2023 + (x-2)^2 = a$$

$$1 + (x-2)^2 = a$$

$$x^2 - 4x + 5 = a$$

$$\frac{2 \pm \sqrt{16-20}}{2} < 0$$



$$x < 0$$

$$a_{00} (1012 - 1011) = 0$$

$$a = (x - 2)^2 = x^2 - 4x + 4$$

$$\left\{ \begin{array}{l} \frac{(y^2 - xy + 4x - 7y + 12) \sqrt{x+5}}{\sqrt{5-x}} = 0 \\ |x| + |y| - a = 0 \end{array} \right. \quad \underline{\underline{2 \text{ pen}}}$$

$$\left\{ \begin{array}{l} y = 4 \\ y = x + 3 \\ x = -5 \\ -5 \leq x \leq 5 \end{array} \right.$$

